In [2]:
```python
import pandas as pd
```

In [46]:
```python
df = pd.read_csv("C:/Users/pratiksha/Downloads/archive (3)/loan_data_set.csv" , sep=",")
```

In [47]:
```python
df
```

Out[47]:

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncom
0	LP001002	Male	No	0	Graduate	No	5849	0
1	LP001003	Male	Yes	1	Graduate	No	4583	1508
2	LP001005	Male	Yes	0	Graduate	Yes	3000	0
3	LP001006	Male	Yes	0	Not Graduate	No	2583	2358
4	LP001008	Male	No	0	Graduate	No	6000	0
...	...	...	...	...	...	...	...	
609	LP002978	Female	No	0	Graduate	No	2900	0
610	LP002979	Male	Yes	3+	Graduate	No	4106	0
611	LP002983	Male	Yes	1	Graduate	No	8072	240
612	LP002984	Male	Yes	2	Graduate	No	7583	0
613	LP002990	Female	No	0	Graduate	Yes	4583	0

614 rows × 13 columns

In [48]:
```python
df.shape
```

Out[48]:
```
(614, 13)
```

In [49]:
```python
df.isnull().sum()
```

Out[49]:
```
Loan_ID               0
Gender               13
Married               3
Dependents           15
Education             0
Self_Employed        32
ApplicantIncome       0
CoapplicantIncome     0
LoanAmount           22
Loan_Amount_Term     14
Credit_History       50
Property_Area         0
Loan_Status           0
dtype: int64
```

In [30]:
```python
mean = df['LoanAmount'].mean()
print(mean)
```

```
146.41216216216216
```

In [31]:
```python
median = df['LoanAmount'].median()
print(median)
```

```
128.0
```

In [55]:
```python
std = df['LoanAmount'].std()
print(std)
```

```
85.58732523570545
```

In [50]:
```python
temp = df[['Gender', 'LoanAmount']].dropna()
```

In [51]:
```python
result = temp.groupby('Gender')['LoanAmount'].agg(['mean', 'median', 'min', 'max', 'std'])
```

In [52]:
```python
print(result)
```

```
              mean  median   min    max        std
Gender
Female  126.697248   113.0   9.0  600.0  79.286460
Male    149.265957   130.0  17.0  650.0  82.810851
```

In [53]:
```python
print(df.groupby('Education')['ApplicantIncome'].agg(['mean','median','min','max','std']))
```

```
                     mean  median  min    max          std
Education
Graduate      5857.433333  4000.0  150  81000  6739.797954
Not Graduate  3777.283582  3357.5  210  18165  2237.081586
```

In [54]:
```python
print(df.groupby('Property_Area')['LoanAmount'].agg(['mean','median','min','max','std']))
```

```
                     mean  median   min    max        std
Property_Area
Rural          152.260116   133.0  40.0  570.0  80.233283
Semiurban      145.504386   127.5  25.0  600.0  81.668261
Urban          142.198953   120.0   9.0  700.0  94.547132
```

In [ ]: